Teoria de Física dos Dispositivos Eletrônicos

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Aula-1

O Método Científico

Matéria e Energia

FCTE **Universidade de Brasília**

Lâmina de Ocam

Occam's Razor

"Lei da Parsimônia"

Vitral
Igreja de Todos os Santos
Surrey, Inglaterra

Princípio proposto pelo frade franciscano e filósofo William of Ockham (1285–1347/49)

"pluralitas non est ponenda sine necessitate"

"plurality should not be posited without necessity."

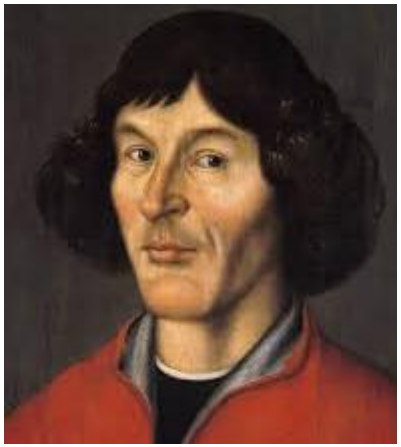
"Non sunt multiplicanda entia sine necessitate."

"Entities are not to be multiplied beyond necessity."

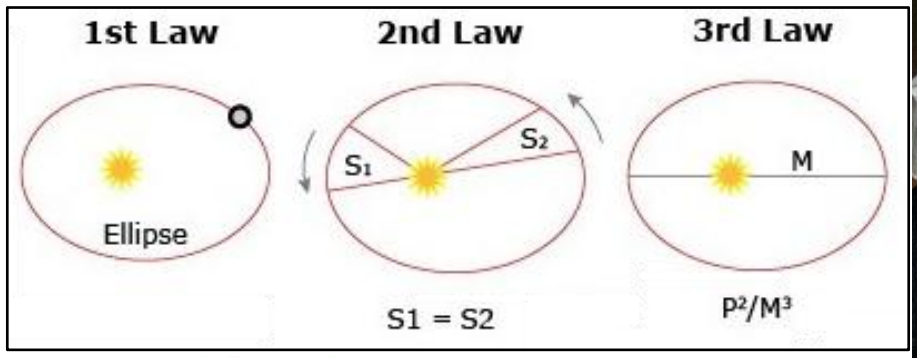
https://www.britannica.com/topic/Occams-razor

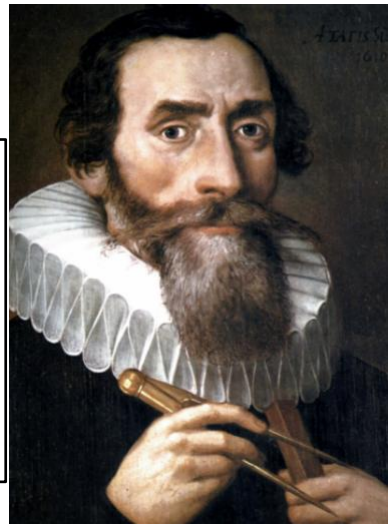
Isaac Newton: **"We are to admit no more causes of natural things than such as are both true and sufficient to explain their appearances."**

Mecânica Celeste

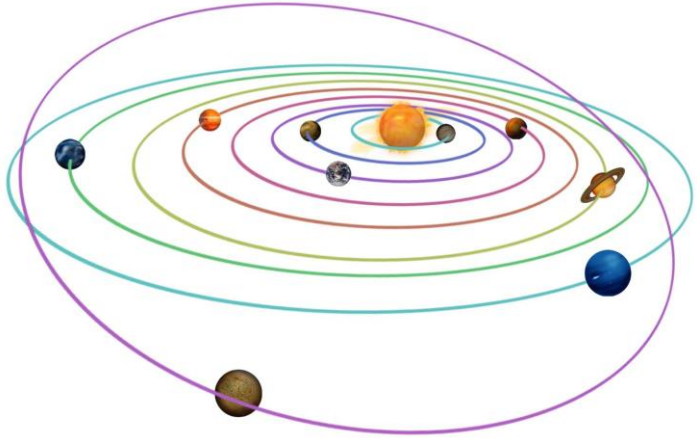


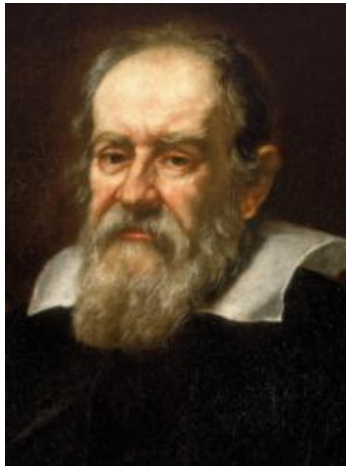
Nicolau Copérnico
1473 - 1543
Heliocentrismo





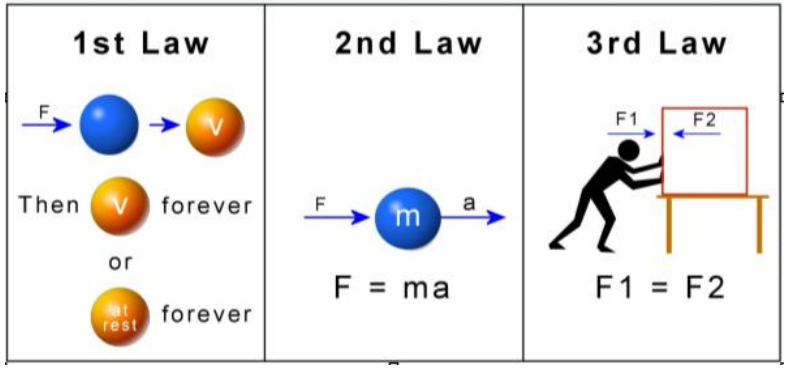
Johannes Kepler
1571 - 1630
Leis das Órbitas

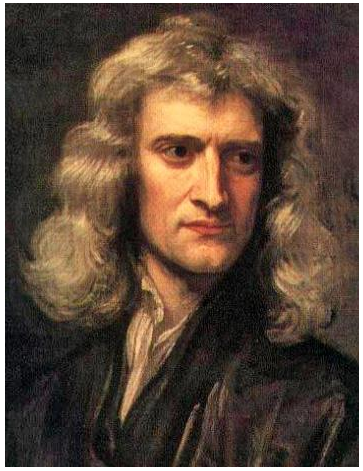




Galileu Galilei
1564 - 1642
Cinemática
Telescópio Astronômico

$$x' = x - vt$$
$$y' = y$$
$$z' = z$$
$$t' = t$$

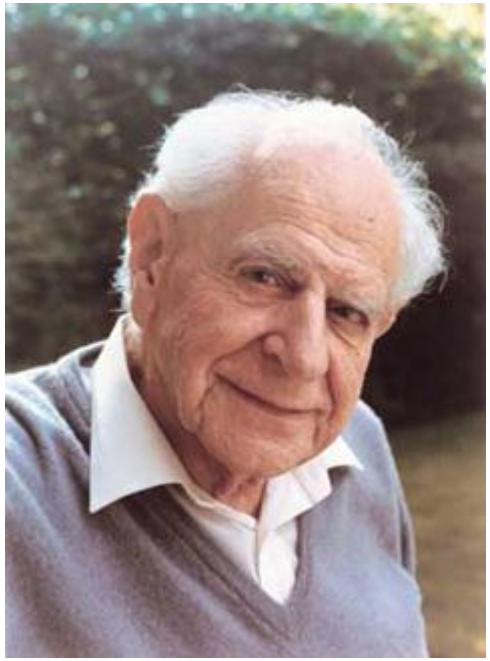




Sir Isaac Newton
1642 - 1727
Mecânica Clássica
Ótica, Cálculo

Fronteira: Ciência e Pseudociência

Falseabilidade (Falsifiability): Propriedade de uma asserção, ideia, hipótese ou teoria, que poder ser mostrada falsa.



Sir Karl Popper
1902 - 1994

Verdadeira Ciência deve poder ser testada e refutada.

Em Ciência devemos buscar a refutação e não a confirmação de ideias.

Livro: "The Logic of Scientific Discovery" (1934)

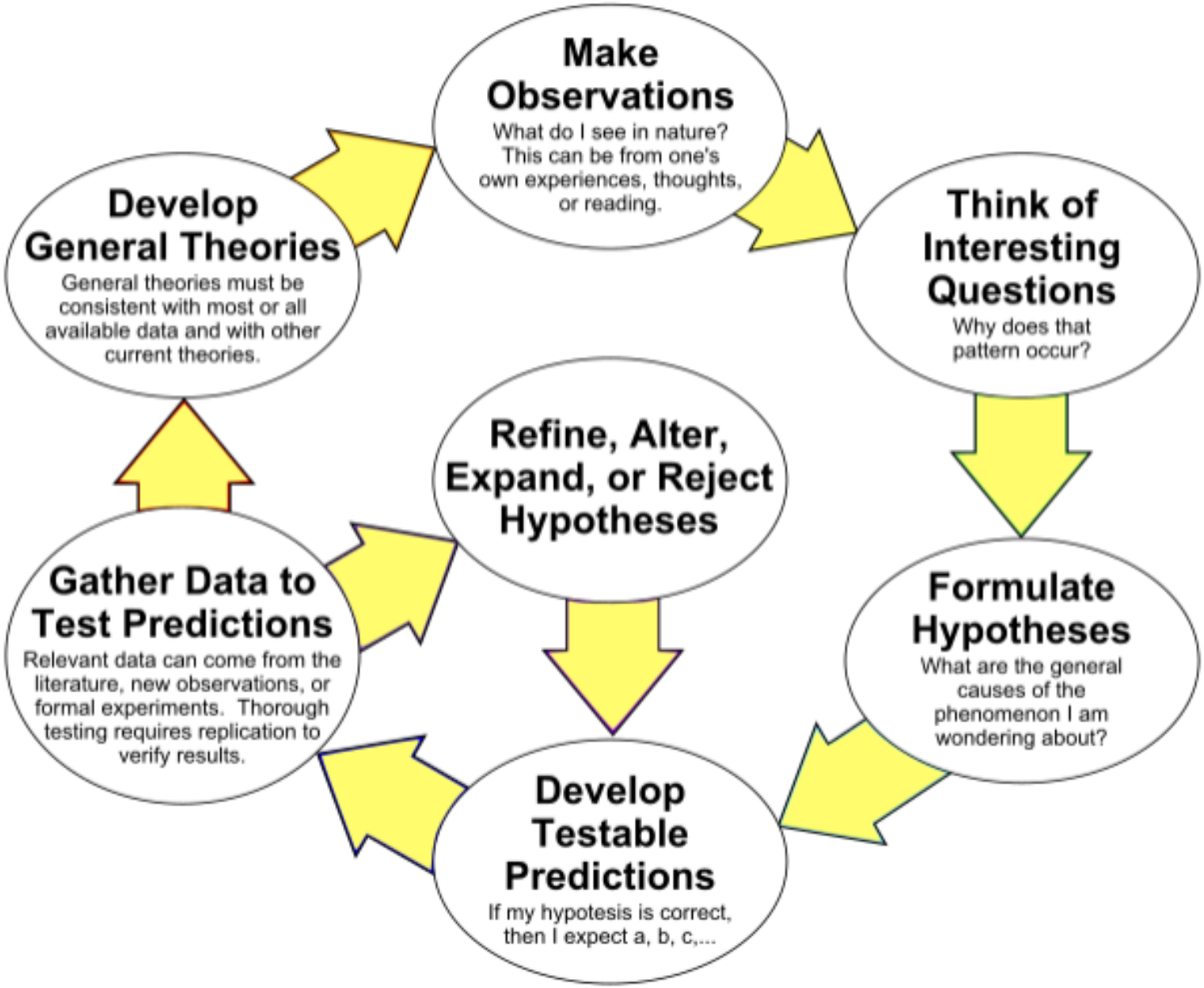
10/08/2023

LK-99: supercondutor revolucionário não funciona em temperatura ambiente

Os cientistas do CMTC sugerem que **o LK-99 apresentou apenas efeitos de diamagnetismo, um termo que explica um efeito de repulsão em ambos os polos magnéticos.** Na Índia, cientistas afirmam que suas réplicas de LK-99 também não funcionaram como supercondutor, apenas com diamagnetismo.

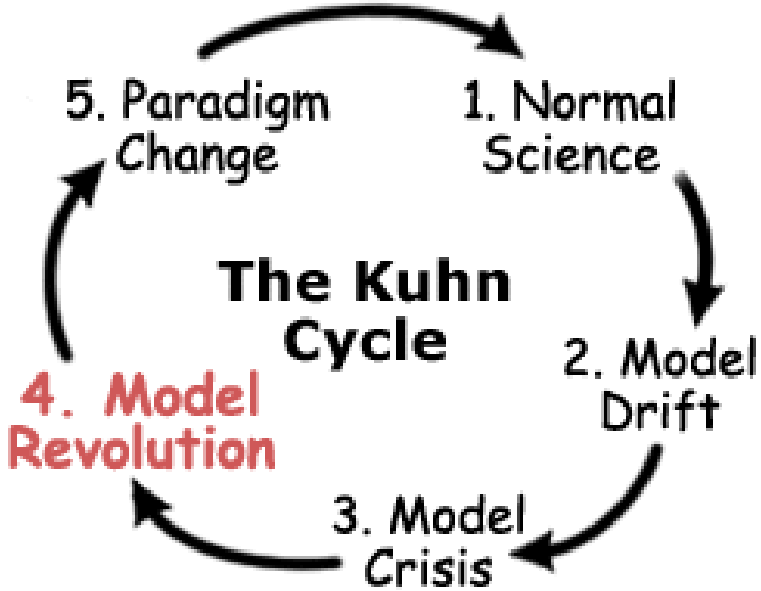
https://www.tecmundo.com.br/ciencia/267426-lk-99-supercondutor-revolucionario-nao-funciona-temperatura-ambiente.htm

The Scientific Method as an Ongoing Process



Garland, Jr., Theodore (2015). "The Scientific Method as an Ongoing Process". U. C. Riverside

The Structure of Scientific Revolutions (1962)



Thomas Samuel Kuhn
(1922 – 1996)

PARADIGMA: "Universally recognized scientific achievements that, for a time, provide model problems and solutions for a community of practitioners".

Fatores que determinam a escolha de teorias:

1. *Accurate* – empirically adequate with experimentation and observation
2. *Consistent* – internally consistent, but also externally consistent with other theories
3. *Broad Scope* – a theory's consequences should extend beyond that which it was initially designed to explain
4. *Simple* – the simplest explanation, principally similar to Occam's razor
5. *Fruitful* – a theory should disclose new phenomena or new relationships among phenomena

O Método Científico

Neil deGrasse Tyson

The Scientific Method

Série: Cosmos

1.Question authority. No idea is true just because someone says so, including me.

2.Think for yourself. Question yourself. Don't believe anything just because you want to. Believing something doesn't make it so.

3.Test ideas by the evidence gained from observation and experiment. If a favorite idea fails a well-designed test, it's wrong. Get over it.

4.Follow the evidence, wherever it leads. If you have no evidence, **reserve judgment**.

5.Remember, you could be wrong. Even the best scientists have been wrong about some things.

Teoria da Relatividade

Relatividade Especial
(1905)

$E = mc^2$

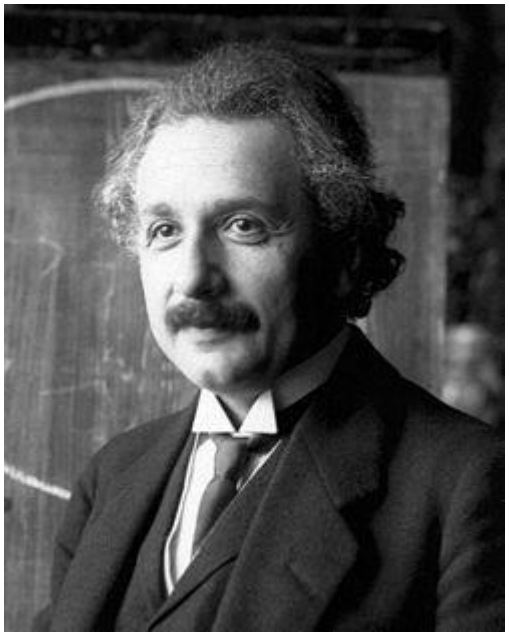
(Matéria e Energia)

$$E^2 = (mc^2)^2 + (pc)^2$$

$$E = \gamma mc^2$$

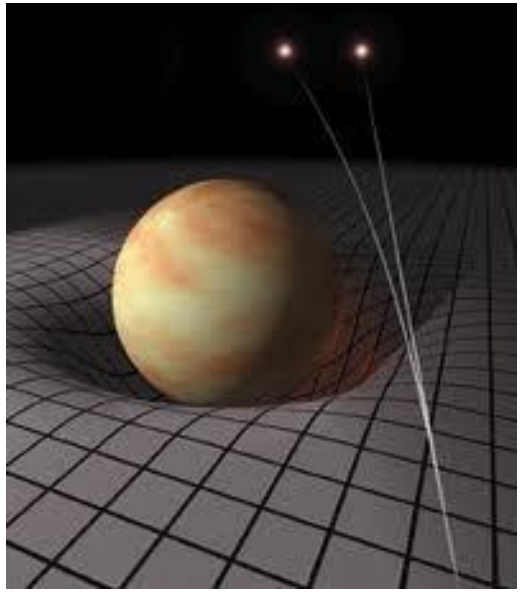
Fóton: $p = E/c$

$$t' = \gamma(t - vx/c^2)$$
$$x' = \gamma(x - vt)$$
$$y' = y$$
$$z' = z$$
$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$



Albert Einstein
1879 - 1955

Relatividade Geral
(1915)



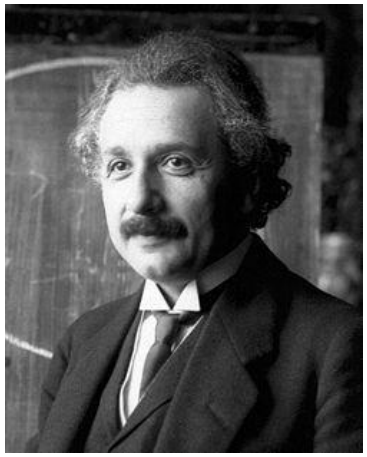
Eq. de Campo:

$$\underbrace{G_{\mu\nu} + \Lambda g_{\mu\nu}}_{\text{tells matter-energy how to curve space-time}} = \underbrace{\frac{8\pi G}{c^4} T_{\mu\nu}}_{\text{tells matter-energy how to move through curved space-time}}$$

$$\kappa = \frac{8\pi G}{c^4} \approx 2.07665(5) \times 10^{-43}\ \mathrm{N}^{-1}$$

Teoria Quântica

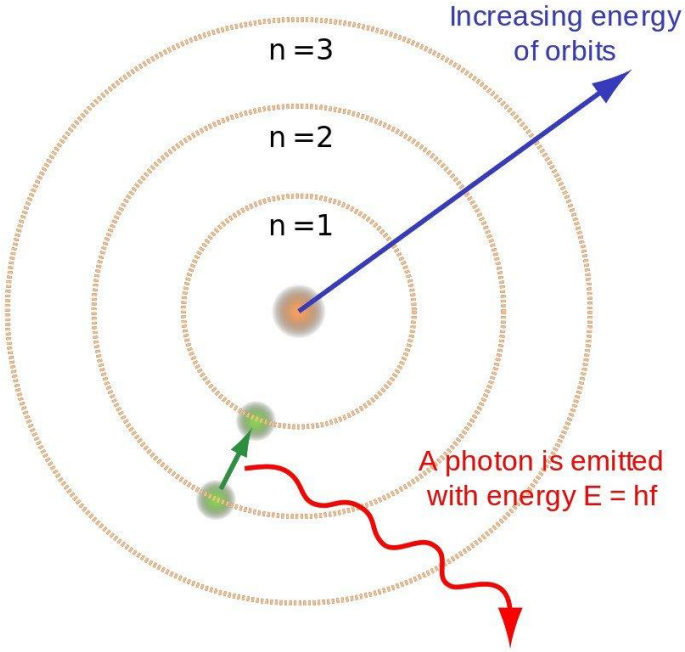
Max Planck
1858 – 1947
Espectro de Radiação
(Corpo Negro)



Albert Einstein
1879 – 1955
Efeito Fotoelétrico

Niels Bohr
1885 – 1962
Estrutura do Átomo

Erwin Schrödinger
1887 – 1961
Equação de Schrödinger



Louis de Broglie
1892 – 1987
Dualidade

Fontes Confiáveis de Informação Acadêmica

Informação ≠ Conhecimento ≠ Entendimento

Prefirir:

- Editoras especializadas que adotem processos de revisão sérios (*peer-review*).

IEEE Xplore®
https://ieeexplore.ieee.org/

nature
https://www.nature.com/

Science
https://www.sciencemag.org/

- Aulas e Tutoriais associados a Universidades e Centros de Pesquisa renomados.

MITOPENCOURSEWARE
MASSACHUSETTS INSTITUTE OF TECHNOLOGY
https://ocw.mit.edu/

Stanford University
https://www.coursera.org/

- *Search Engines* Especializados

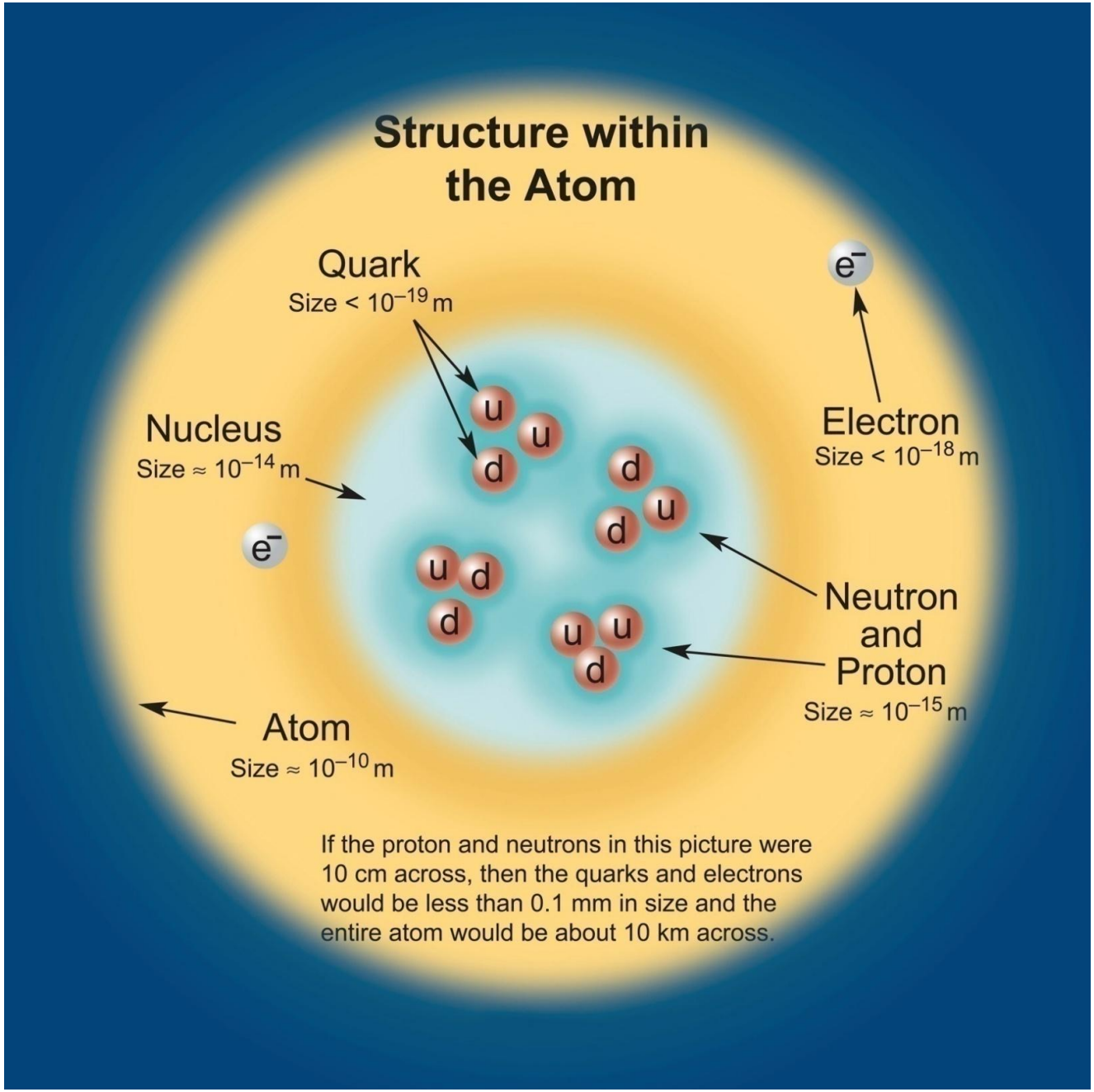
Google Acadêmico

Evitar:

- Traduções de baixa qualidade, feitas por leigos.
- Conteúdos na Internet não assinados, ou produzidos por amadores.
- Linguagem obscura, ou excessivamente coloquial e informal.

Adotar sempre uma postura crítica sobre qualquer conteúdo.
Buscar sempre outras fontes para validar as informações coletadas.

Matéria



http://www.cpepweb.org/

Hádrons

Baryons qqq and Antibaryons $\bar{q}\bar{q}\bar{q}$

Baryons are fermionic hadrons.
There are about 120 types of baryons.

Symbol	Name	Quark content	Electric charge	Mass GeV/c^2	Spin
p	proton	uud	1	0.938	1/2
$\bar{p}$	anti-proton	$\bar{u}\bar{u}\bar{d}$	–1	0.938	1/2
n	neutron	udd	0	0.940	1/2
Λ	lambda	uds	0	1.116	1/2
Ω^-	omega	sss	–1	1.672	3/2

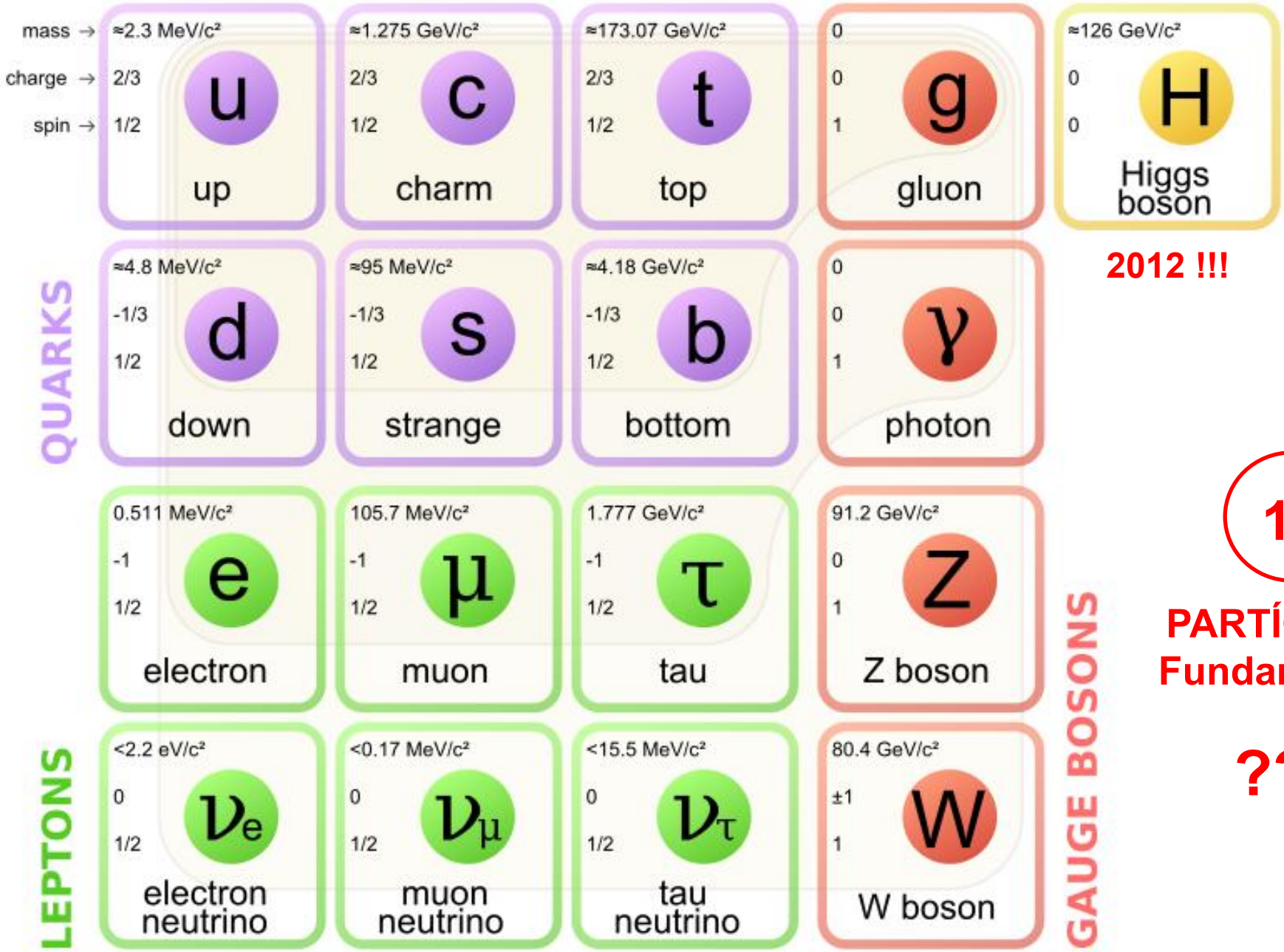
Hádrons

Mesons $q\bar{q}$

Mesons are bosonic hadrons.
There are about 140 types of mesons.

Symbol	Name	Quark content	Electric charge	Mass GeV/c^2	Spin
π^+	pion	$u\bar{d}$	+1	0.140	0
K^-	kaon	$s\bar{u}$	–1	0.494	0
ρ^+	rho	$u\bar{d}$	+1	0.770	1
B^0	B-zero	$d\bar{b}$	0	5.279	0
η_c	eta-c	$c\bar{c}$	0	2 .980	0

Partículas Elementares do Modelo Padrão



2012 !!!

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PARTÍCULAS Fundamentais

???

Standard Model of Elementary Particles

	three generations of matter (elementary fermions)			three generations of antimatter (elementary antifermions)			interactions / force carriers (elementary bosons)	
	I	II	III	I	II	III		
mass / charge / spin	≃2.2 MeV/c² / ⅔ / ½	≃1.28 GeV/c² / ⅔ / ½	≃173.1 GeV/c² / ⅔ / ½	≃2.2 MeV/c² / −⅔ / ½	≃1.28 GeV/c² / −⅔ / ½	≃173.1 GeV/c² / −⅔ / ½	0 / 0 / 1	≃124.97 GeV/c² / 0 / 0
QUARKS	u up	c charm	t top	ū antiup	c̄ anticharm	t̄ antitop	g gluon	H higgs
	≃4.7 MeV/c² / −⅓ / ½	≃96 MeV/c² / −⅓ / ½	≃4.18 GeV/c² / −⅓ / ½	≃4.7 MeV/c² / ⅓ / ½	≃96 MeV/c² / ⅓ / ½	≃4.18 GeV/c² / ⅓ / ½	0 / 0 / 1	
	d down	s strange	b bottom	d̄ antidown	s̄ antistrange	b̄ antibottom	γ photon	
LEPTONS	≃0.511 MeV/c² / −1 / ½	≃105.66 MeV/c² / −1 / ½	≃1.7768 GeV/c² / −1 / ½	≃0.511 MeV/c² / 1 / ½	≃105.66 MeV/c² / 1 / ½	≃1.7768 GeV/c² / 1 / ½	≃91.19 GeV/c² / 0 / 1	
	e electron	μ muon	τ tau	e^+ positron	$\bar{\mu}$ antimuon	$\bar{\tau}$ antitau	Z Z⁰ boson	
	<2.2 eV/c² / 0 / ½	<0.17 MeV/c² / 0 / ½	<18.2 MeV/c² / 0 / ½	<2.2 eV/c² / 0 / ½	<0.17 MeV/c² / 0 / ½	<18.2 MeV/c² / 0 / ½	≃80.39 GeV/c² / 1 / 1	≃80.39 GeV/c² / −1 / 1
	ν_e electron neutrino	ν_μ muon neutrino	ν_τ tau neutrino	$\bar{\nu}_e$ electron antineutrino	$\bar{\nu}_\mu$ muon antineutrino	$\bar{\nu}_\tau$ tau antineutrino	W^+ W^+ boson	W^- W^- boson

GAUGE BOSONS
VECTOR BOSONS
SCALAR BOSONS

30 **PARTÍCULAS Fundamentais** **???**

Partículas Elementares do Modelo Padrão

The 17 named fundamental particles of the Standard Model

12 fermions			5 bosons
up quark	charm quark	top quark	gluon
down quark	strange quark	bottom quark	photon
electron	muon	tau	W boson
electron neutrino	muon neutrino	tau neutrino	Z boson
			higgs boson

http://physics.info/standard/practice.shtml

Férmions de Matéria (24)

Quarks come in 6 flavors and 3 colors, which gives us 18 unique quarks.

18 quarks		
red up quark	red charm quark	red top quark
green up quark	green charm quark	green top quark
blue up quark	blue charm quark	blue top quark
red down quark	red strange quark	red bottom quark
green down quark	green strange quark	green bottom quark
blue down quark	blue strange quark	blue bottom quark

The leptons come in 6 flavors, 3 of which are neutrinos, but none of which are colored. There is nothing to multiply here. Just add 6 leptons to the 18 quarks to get 24 matter fermions.

6 leptons		
electron	muon	tau
electron neutrino	muon neutrino	tau neutrino

Férmions de Antimatéria (24)

18 antiquarks		
antired antiup quark	antired anticharm quark	antired antitop quark
antigreen antiup quark	antigreen anticharm quark	antigreen antitop quark
antiblue antiup quark	antiblue anticharm quark	antiblue antitop quark
antired antidown quark	antired antistrange quark	antired antibottom quark
antigreen antidown quark	antigreen antistrange quark	antigreen antibottom quark
antiblue antidown quark	antiblue antistrange quark	antiblue antibottom quark
6 antileptons		
antielectron*	antimuon	antitau
electron antineutrino	muon antineutrino	tau antineutrino

* also known as the positron

Bósons (13)

Gluons come in 8 color combinations. Here's one way to write them out.

8 gluons		
$\frac{r\bar{b} + b\bar{r}}{\sqrt{2}}$	$\frac{-i\,(r\bar{b} - b\bar{r})}{\sqrt{2}}$	$\frac{r\bar{r} - b\bar{b}}{\sqrt{2}}$
$\frac{r\bar{g} + g\bar{r}}{\sqrt{2}}$	$\frac{-i\,(r\bar{g} - g\bar{r})}{\sqrt{2}}$	
$\frac{g\bar{b} + b\bar{g}}{\sqrt{2}}$	$\frac{-i\,(g\bar{b} - b\bar{g})}{\sqrt{2}}$	$\frac{r\bar{r} + g\bar{g} - 2b\bar{b}}{\sqrt{6}}$

4 electroweak bosons			
photon	W^+ boson	W^- boson	Z^0 boson

1 higgs bosons
higgs boson

TOTAL: 61 PARTÍCULAS FUNDAMENTAIS IDENTIFICADAS

Elementary Particles

	Types	Generations	Antiparticle	Colours	Total
Quarks	2	3	Pair	3	36
Leptons			Pair	None	12
Gluons	1	1	Own	8	8
Photon			Own	None	1
Z Boson			Own		1
W Boson			Pair		2
Higgs			Own		1
Total number of (known) elementary particles:					61

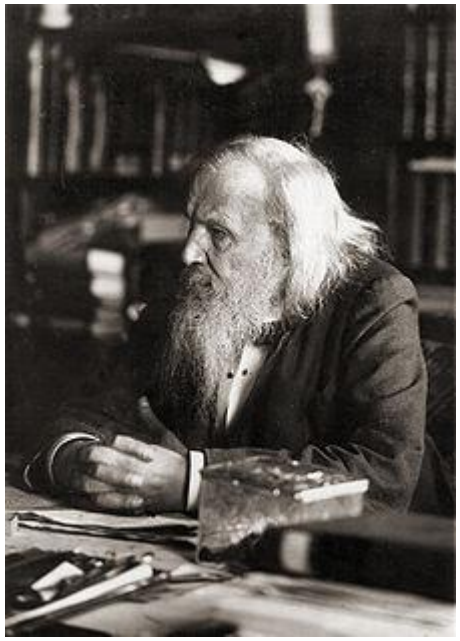
Quatro Forças Fundamentais

Properties of the Interactions

The strengths of the interactions (forces) are shown relative to the strength of the electromagnetic force for two u quarks separated by the specified distances.

Property	Gravitational Interaction	Weak Interaction (Electroweak)	Electromagnetic Interaction	Strong Interaction
Acts on:	Mass – Energy	Flavor	Electric Charge	Color Charge
Particles experiencing:	All	Quarks, Leptons	Electrically Charged	Quarks, Gluons
Particles mediating:	Graviton (not yet observed)	W^+ W^- Z^0	γ	Gluons
Strength at 10^{-18} m	10^{-41}	0.8	1	25
Strength at 3×10^{-17} m	10^{-41}	10^{-4}	1	60

http://www.cpepweb.org/

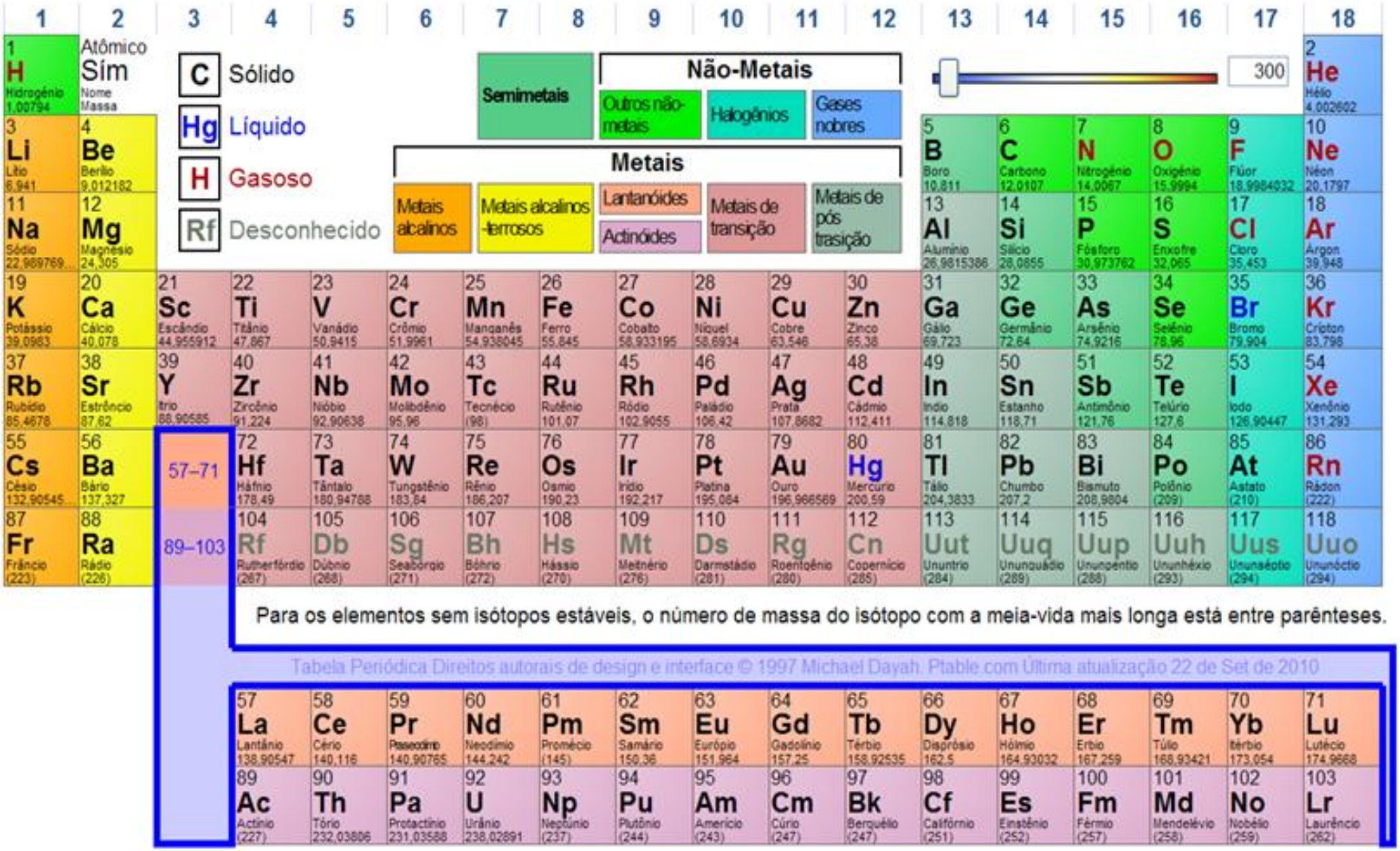


Dmitri Ivanovich Mendeleev
(1834 – 1907)

Reihen	Gruppo I. — R^2O	Gruppo II. — RO	Gruppo III. — R^2O^3	Gruppo IV. RH^4 RO^2	Gruppo V. RH^3 R^2O^5	Gruppo VI. RH^2 RO^3	Gruppo VII. RH R^2O^7	Gruppo VIII. — RO^4
1	H=1							
2	Li=7	Be=9,4	B=11	C=12	N=14	O=16	F=19	
3	Na=23	Mg=24	Al=27,3	Si=28	P=31	S=32	Cl=35,5	
4	K=39	Ca=40	—=44	Ti=48	V=51	Cr=52	Mn=55	Fe=56, Co=59, Ni=59, Cu=63.
5	(Cu=63)	Zn=65	—=68	—=72	As=75	Se=78	Br=80	
6	Rb=85	Sr=87	?Yt=88	Zr=90	Nb=94	Mo=96	—=100	Ru=104, Rh=104, Pd=106, Ag=108.
7	(Ag=108)	Cd=112	In=113	Sn=118	Sb=122	Te=125	J=127	
8	Cs=133	Ba=137	?Di=138	?Ce=140	—	—	—	— — — —
9	(—)	—	—	—	—	—	—	
10	—	—	?Er=178	?La=180	Ta=182	W=184	—	Os=195, Ir=197, Pt=198, Au=199.
11	(Au=199)	Hg=200	Tl=204	Pb=207	Bi=208	—	—	
12	—	—	—	Th=231	—	U=240	—	— — — —

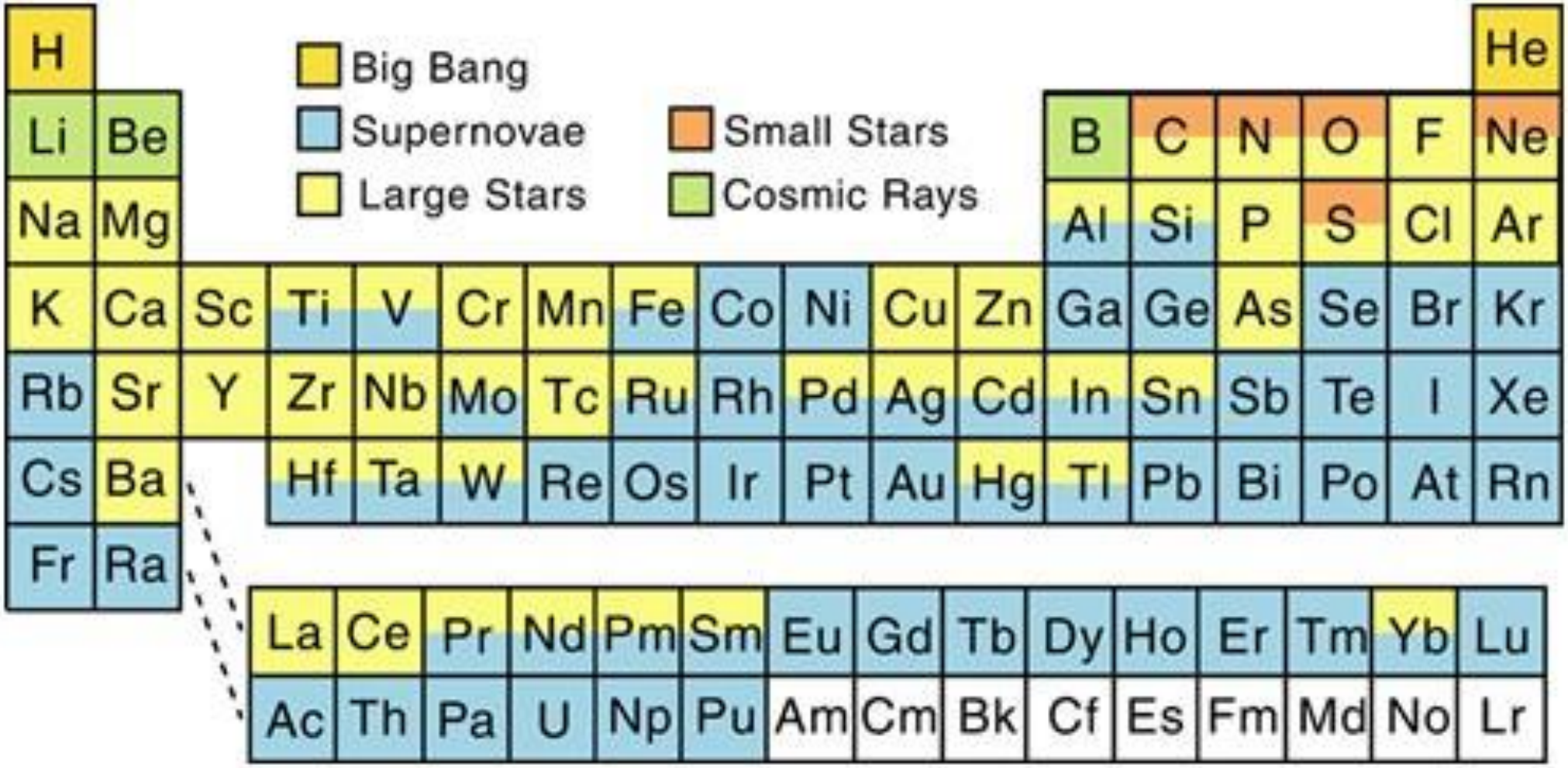
Tabela Periódica - 1871

Tabela Periódica Moderna



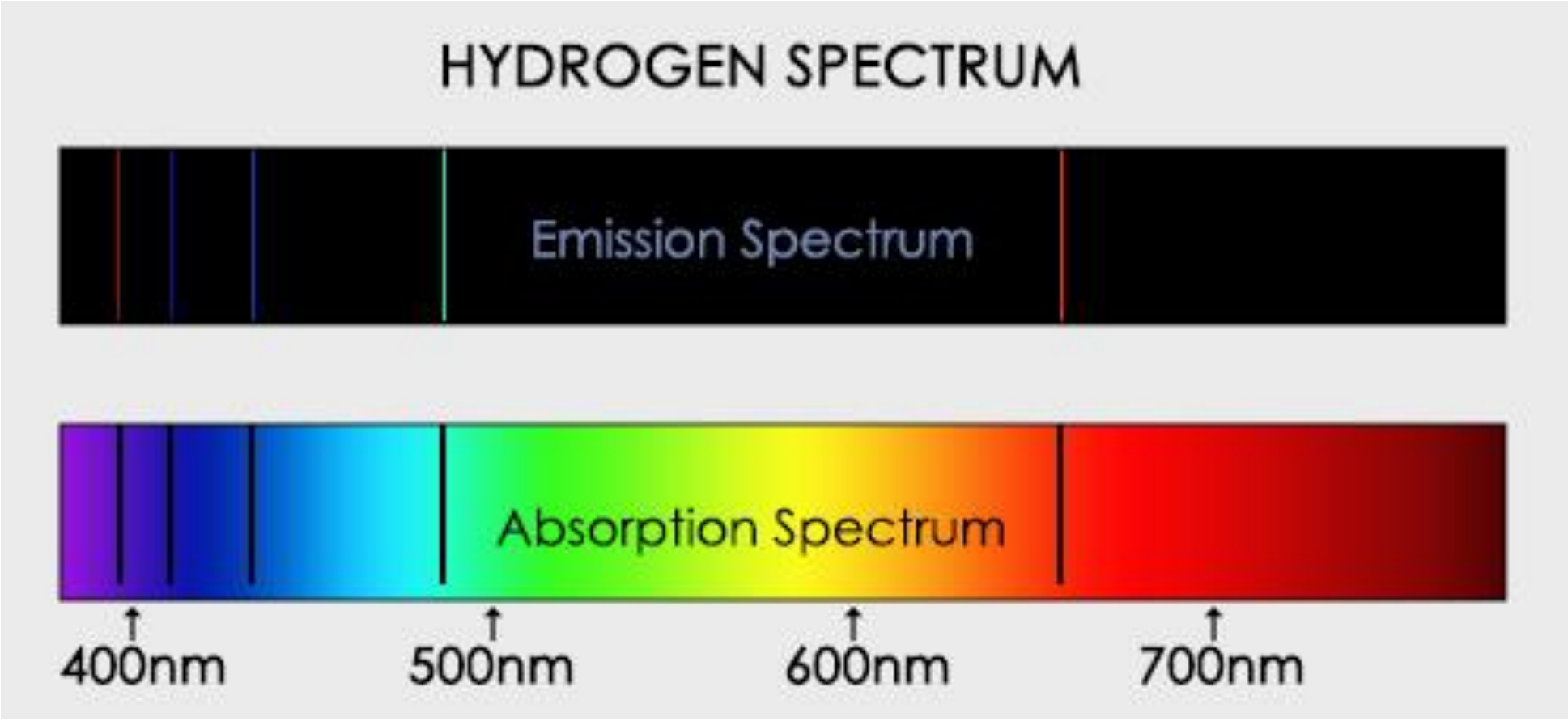
http://www.ptable.com/

De onde surgiram os Elementos?



https://www.universetoday.com/70646/astronomy-without-a-telescope-alchemy-by-supernova/

Espectro do Hidrogênio



Estados da Matéria

A existência de um quinto estado da matéria só foi comprovada em laboratório em 1995. Ele havia sido previsto em 1924 pelo físico alemão Albert Einstein e pelo matemático indiano Satyendra Nath Bose

ESTADO DE PLASMA

A temperaturas altíssimas, da ordem de 10.000 °C, os átomos possuem carga elétrica. Movem-se caoticamente, espalhando-se para todas as direções, a cerca de 15.000 km/h. É o estado da matéria das estrelas

ESTADO GASOSO

Abaixo de 1.000 °C, as partículas começam a frear e a neutralizar sua carga elétrica. Num gás à temperatura ambiente, os átomos viajam com velocidade média de 1.500 km/h, também em qualquer direção

ESTADO LÍQUIDO

À temperatura ambiente, os átomos da maior parte das substâncias viajam ainda desordenadamente. Percorrem todo o volume que lhes é dado, como o de um copo. A velocidade média das partículas é bem menor: 90 km/h

ESTADO SÓLIDO

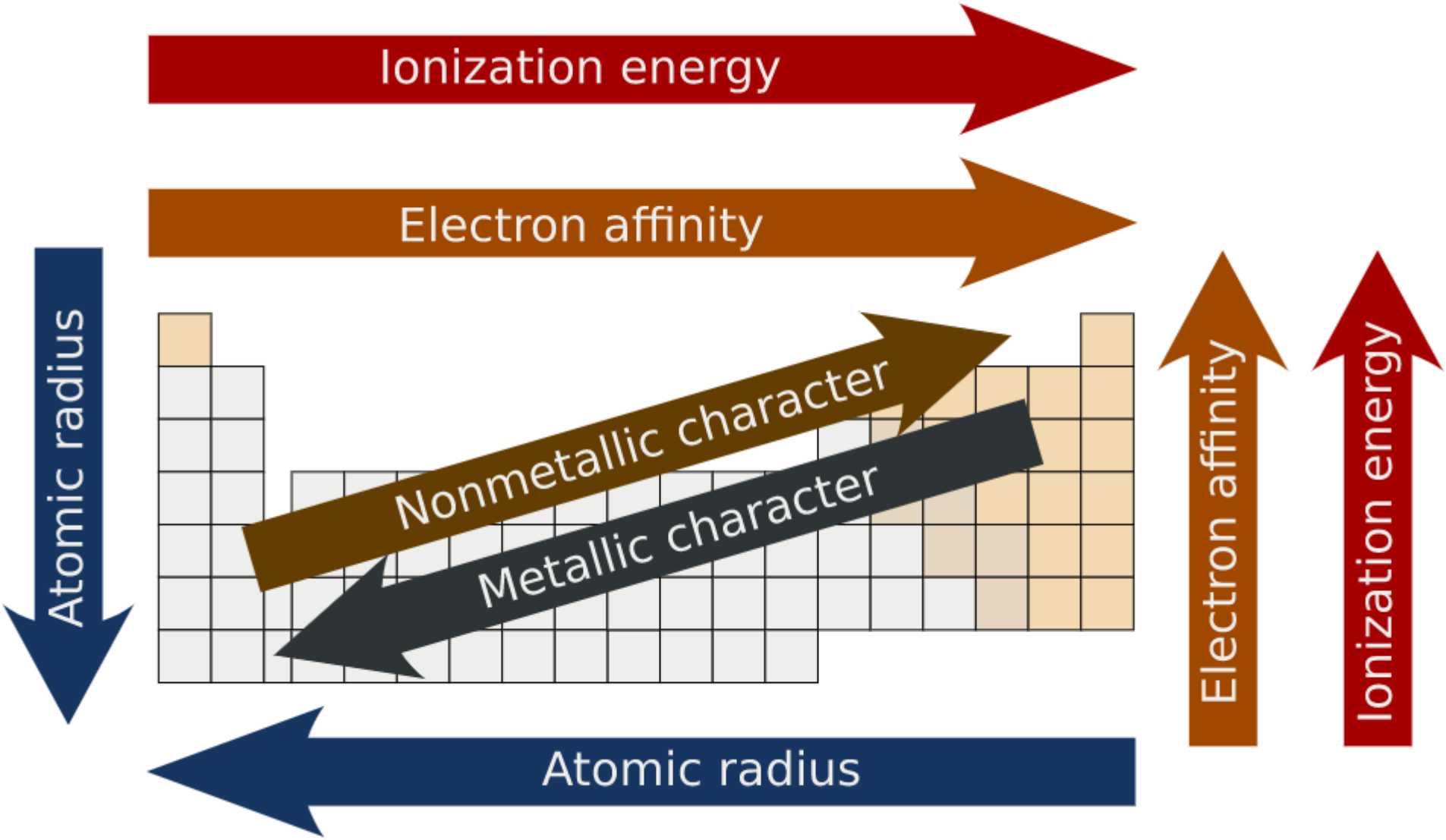
Quando a temperatura cai mais, os átomos se unem firmemente. Mas ainda dançam. Na água, a 0 °C, os átomos andam com a mesma velocidade de 90 km/h, mas já não caminham por todo o volume. Apenas vibram em torno de um ponto

NÃO EXISTE NA NATUREZA

QUINTO ESTADO

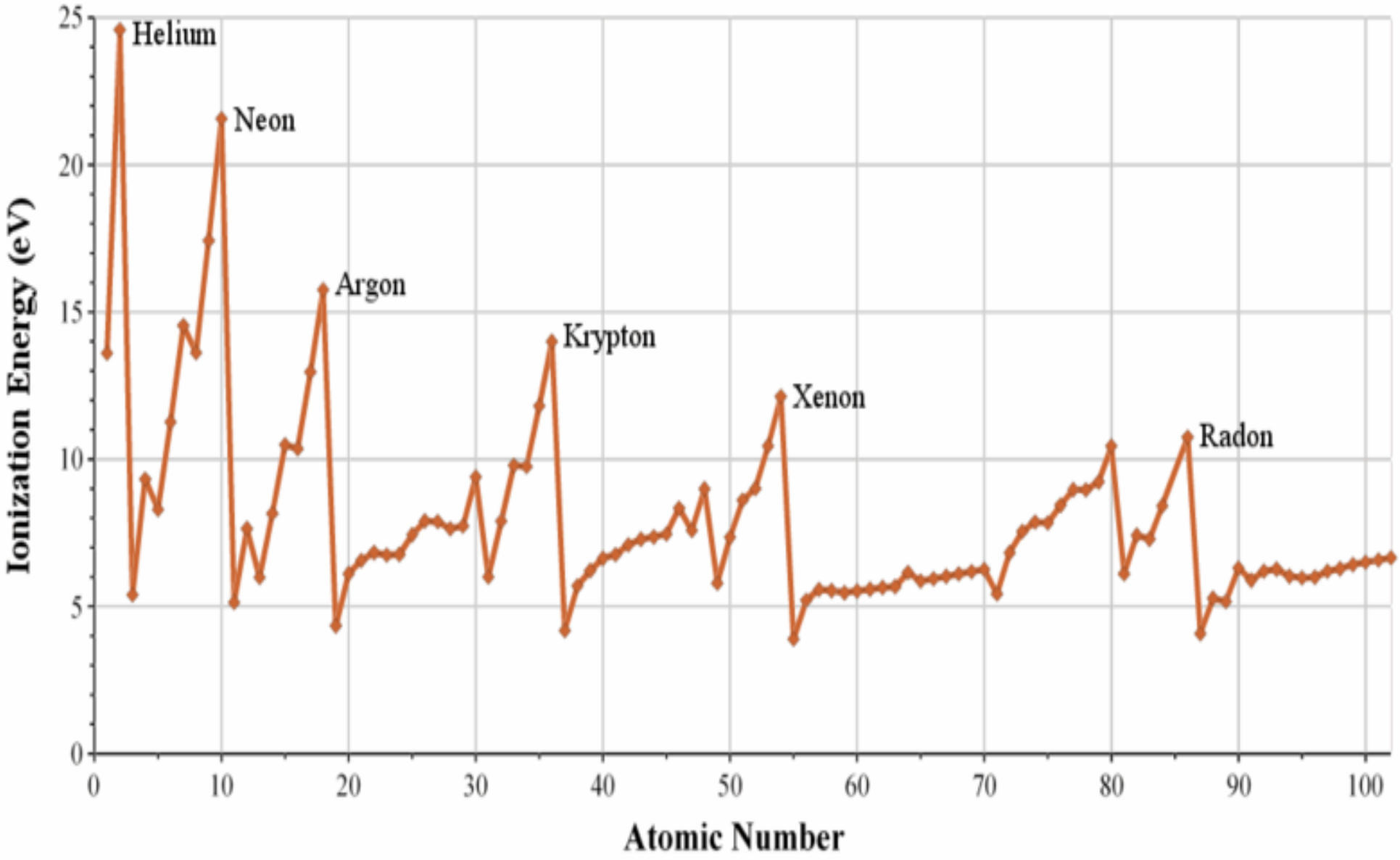
No condensado Bose-Einstein, os átomos estão a uma temperatura muito próxima do zero absoluto (-273 °C). As partículas vibram como um corpo único, numa velocidade tão baixa que é impossível medi-la em laboratório

Tabela Periódica

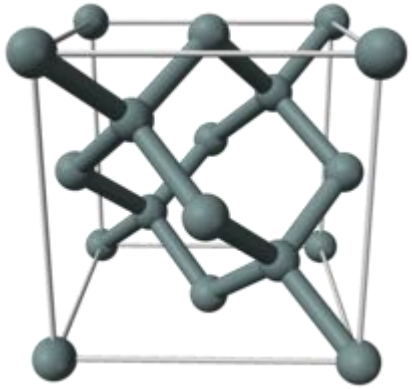


https://courses.lumenlearning.com/boundless-chemistry/chapter/periodic-trends/

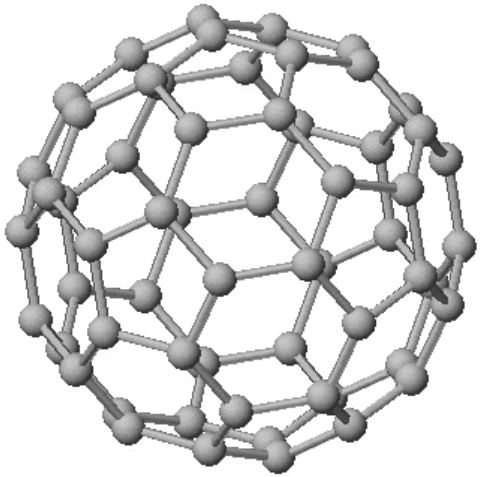
Energia de Ionização



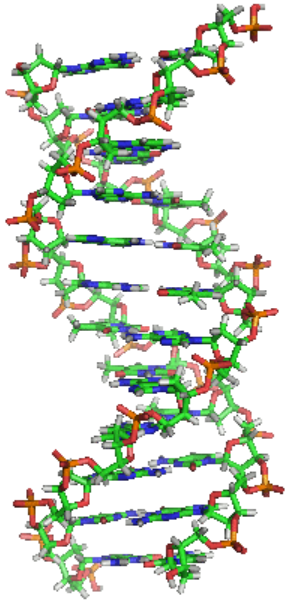
Organização da Matéria Sólida



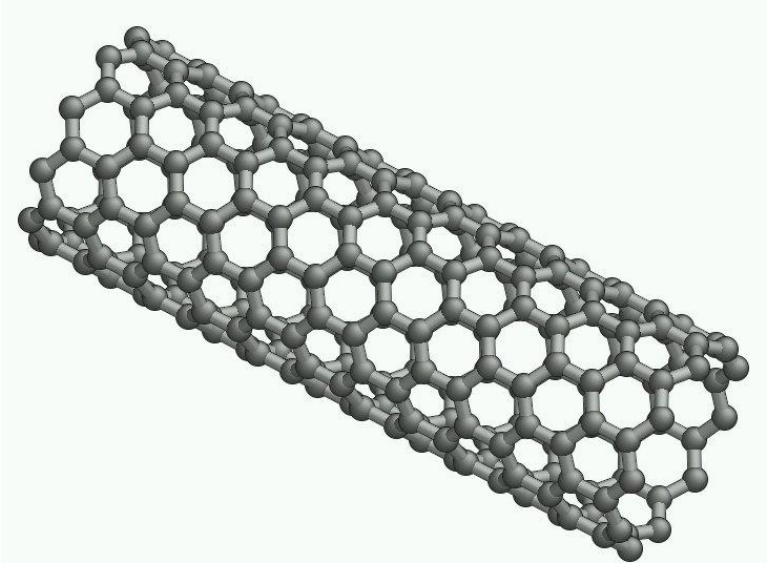
Célula Unitária – Diamante



BuckyBall – C60



DNA



Nanotubo de Carbono

Organização da Matéria

Partículas Fundamentais

Átomos

Moléculas

Matéria

Inorgânica

Crosta Terrestre, Oceanos e Atmosfera

Estratos de Depósitos Minerais (cristais naturais)

Orgânica

RNA, DNA, Proteínas

Células (~3 Bilhões de anos)

Seres Multicelulares (Epigenética)

Espécies (Presas e Predadores)

Sociedades (Cooperação/Competição)

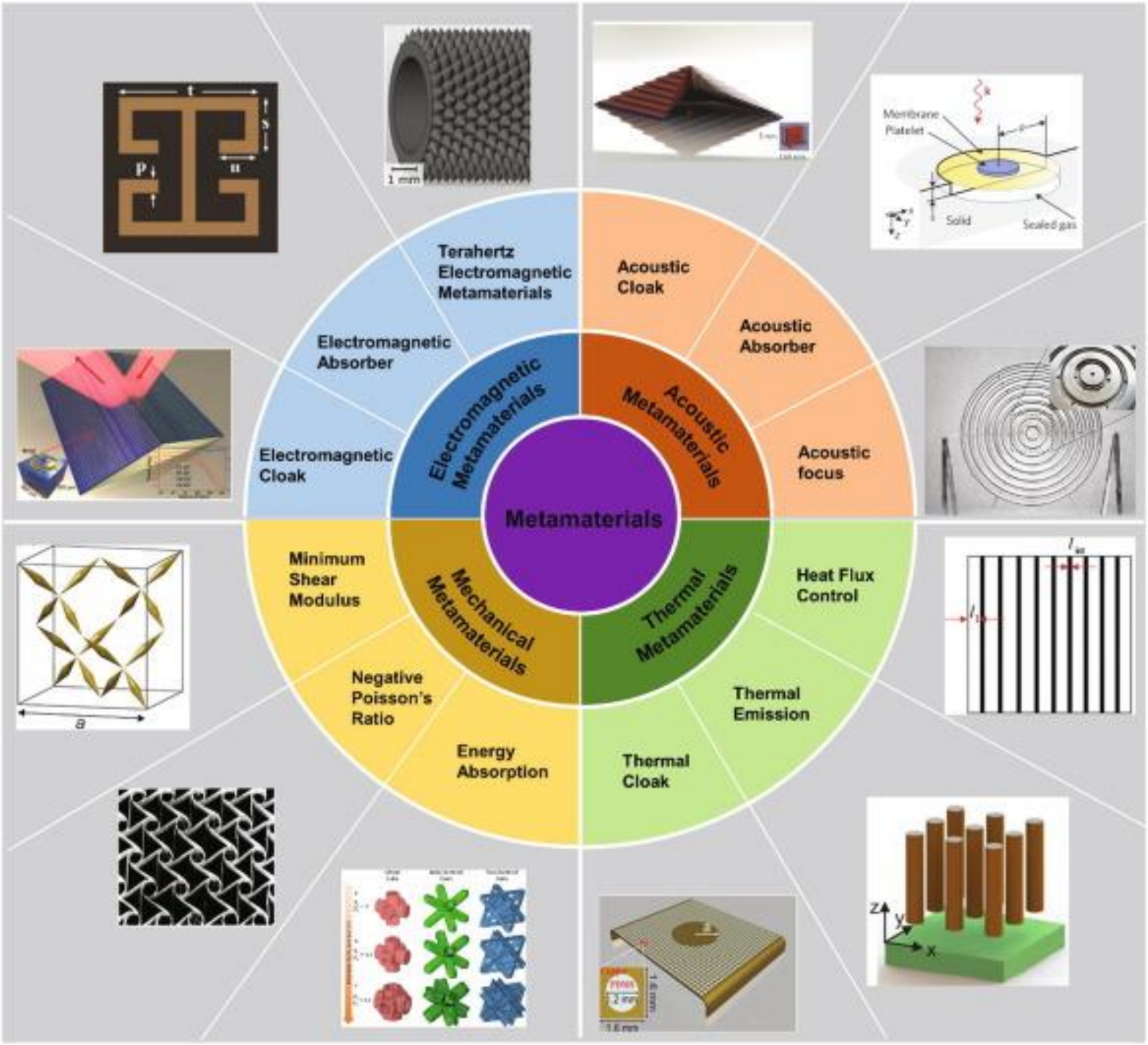
?

Novos tipos de Matéria

VIDA

What is Life?
Erwin Schrödinger
1944
"aperiodic crystal"

Metamateriais



Física de Dispositivos Eletrônicos

Matéria Comum: Átomos
- Elétrons (Lacunas)
- Núcleos Atômicos
  - Prótons
  - Nêutrons

Forças:
- Eletromagnética – Fótons
- Gravidade

Campos Eletromagnéticos x Matéria

Equações do Eletromagnetismo Clássico

Gauss: $\begin{cases} \nabla \cdot \mathbf{D} = \rho \\ \nabla \cdot \mathbf{B} = 0 \end{cases}$

Faraday: $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$

Ampère: $\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$

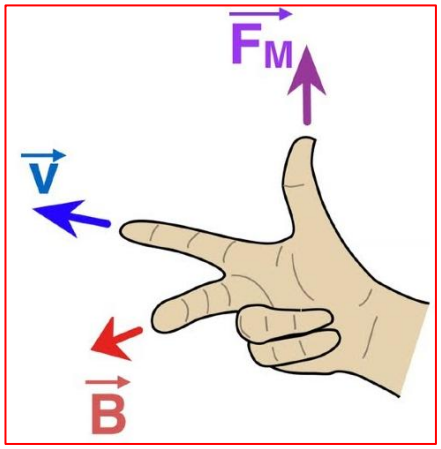
Continuidade: $\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}$

$$\left(\nabla^2 - \mu\epsilon \frac{\partial^2}{\partial t^2}\right)\mathbf{E} = 0$$

$$\left(\nabla^2 - \mu\epsilon \frac{\partial^2}{\partial t^2}\right)\mathbf{B} = 0$$

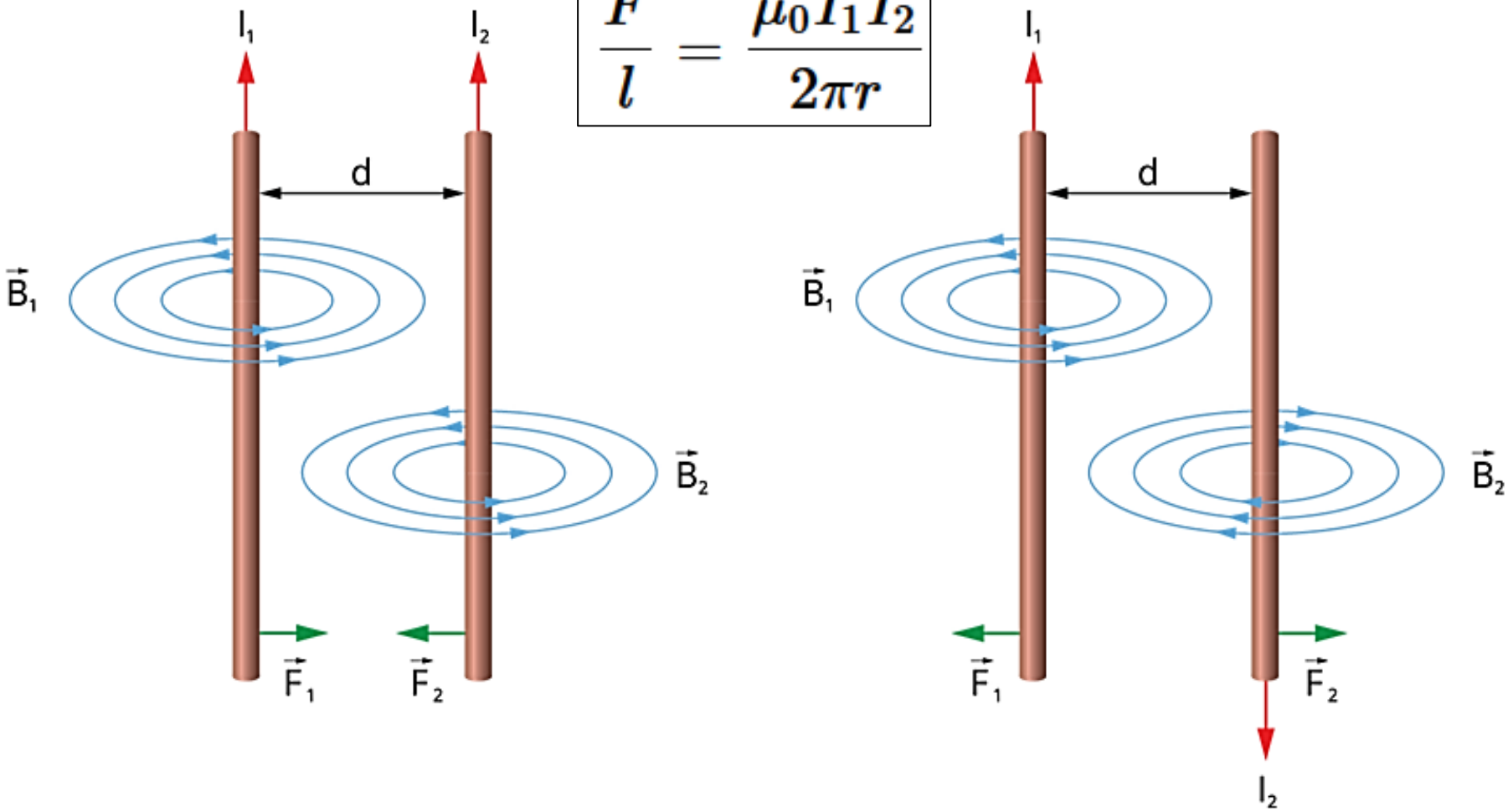
$$c = \frac{1}{\sqrt{\mu\epsilon}}$$

Lorentz: $\vec{F} = q\left[\vec{E} + \vec{v} \times \vec{B}\right]$



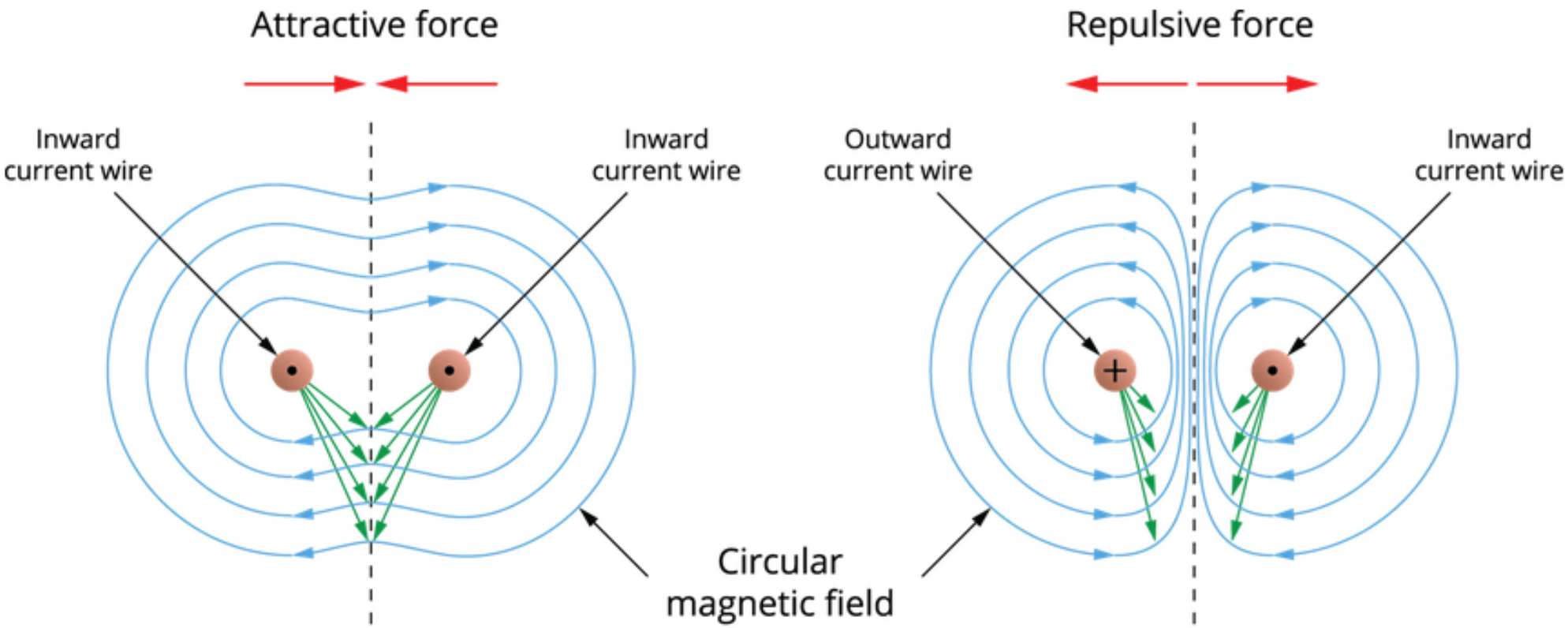
Força entre dois Fios conduzindo Corrente

$$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi r}$$

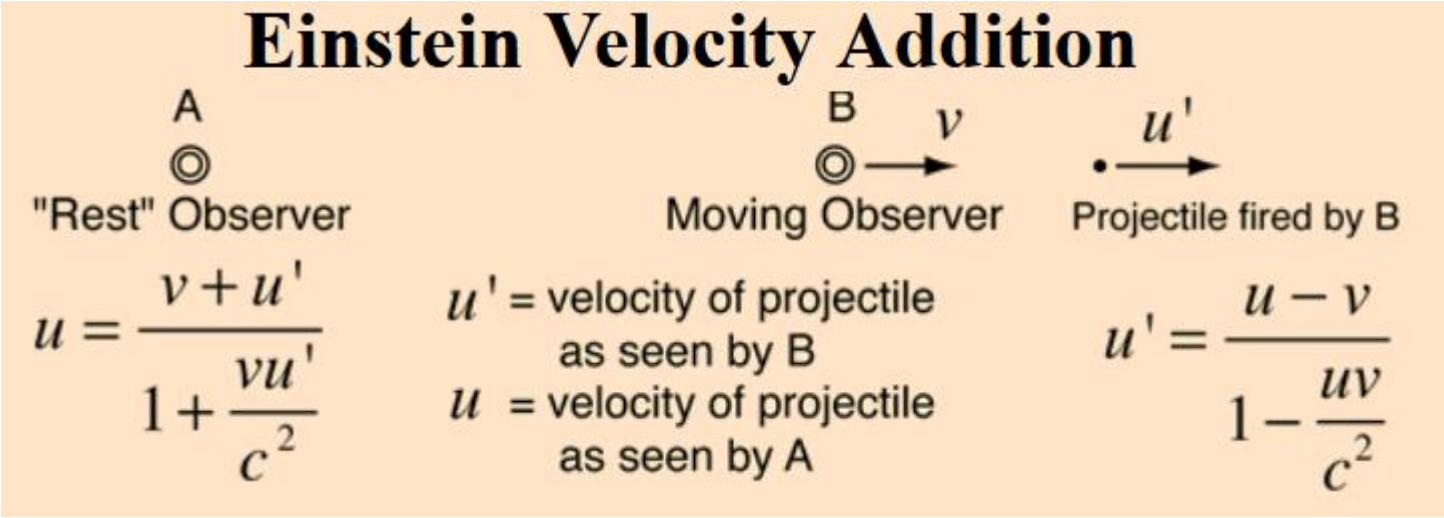


Attractive and repulsive force on a current-carrying conductor

https://www.yaclass.in/p/science-state-board/class-9/magnetism-and-electromagnetism-14262/the-properties-effects-and-applications-10216/re-0082f3b9-538b-441b-80db-608febc7009b



https://www.yaclass.in/p/science-state-board/class-9/magnetism-and-electromagnetism-14262/the-properties-effects-and-applications-10216/re-0082f3b9-538b-441b-80db-608febc7009b



Velocidades Paralelas

http://hyperphysics.phy-astr.gsu.edu/hbase/Relativ/einvel.html

Fator de Lorentz: $$\gamma(v) = \frac{1}{\sqrt{1 - v^2/c^2}}$$

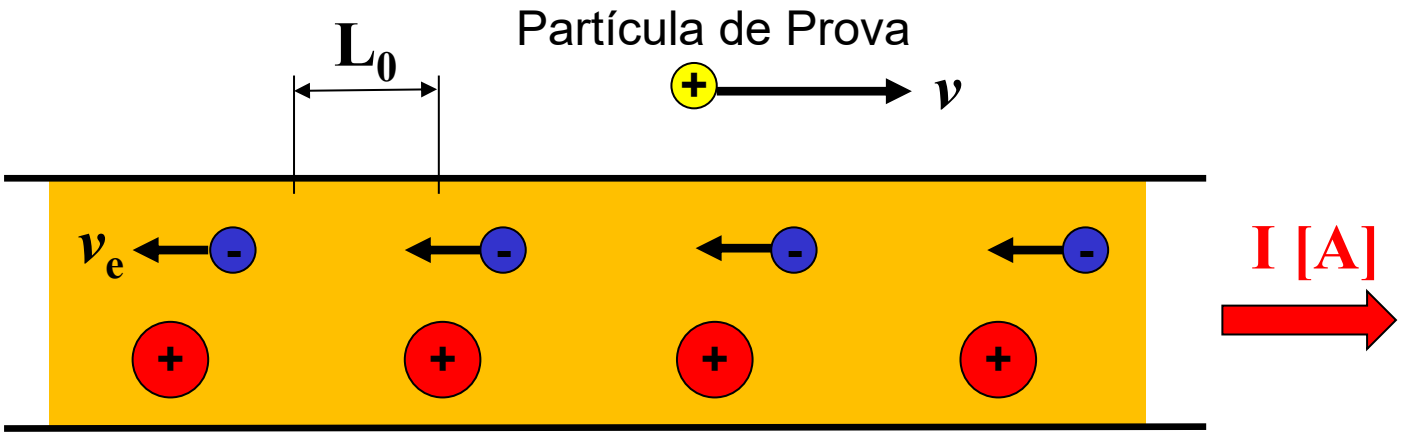
Contração de Comprimento: $$L = \frac{L_0}{\gamma(v)}$$

Massa Relativística: $$m_{rel} = \gamma(v)\, m_0$$

Dilatação do Tempo: $$\Delta'_t = \gamma(v)\, \Delta_t$$

Densidade Relativística de Cargas em Condutor

Velocidades Paralelas



Fio condutor metálico infinito

$$u = (v + v_e)/(1 + (vv_e)/c^2)$$

Velocidade dos elétrons observada pela Partícula de Prova

Densidades Lineares de Cargas [C/m]: $\lambda_+ = \dfrac{Q_+}{L_0}$ $\quad \lambda_- = \dfrac{Q_-}{L_0}$

$$\lambda_+ - \lambda_- = 0$$

$$\lambda_{rel} = \gamma(v)\lambda_+ - \frac{\gamma(u)}{\gamma(v_e)}\lambda_- = \frac{Q_+}{L_0/\gamma(v)} - \frac{Q_-}{L_0\gamma(v_e)/\gamma(u)}$$

Densidade de Carga observada pela Partícula de Prova

Contração

$$u = (v + v_e)/(1 + (vv_e)/c^2) \qquad \gamma(u) = \frac{1}{\sqrt{1 - u^2/c^2}}$$

$$\gamma(u) = \left(1 + \frac{vv_e}{c^2}\right)\left\{\left(1 + \frac{vv_e}{c^2}\right)^2 - \left(\frac{v+v_e}{c^2}\right)\right\}^{1/2}$$

$$\gamma(u) = \left(1 + \frac{vv_e}{c^2}\right)\gamma(v)\,\gamma(v_e)$$

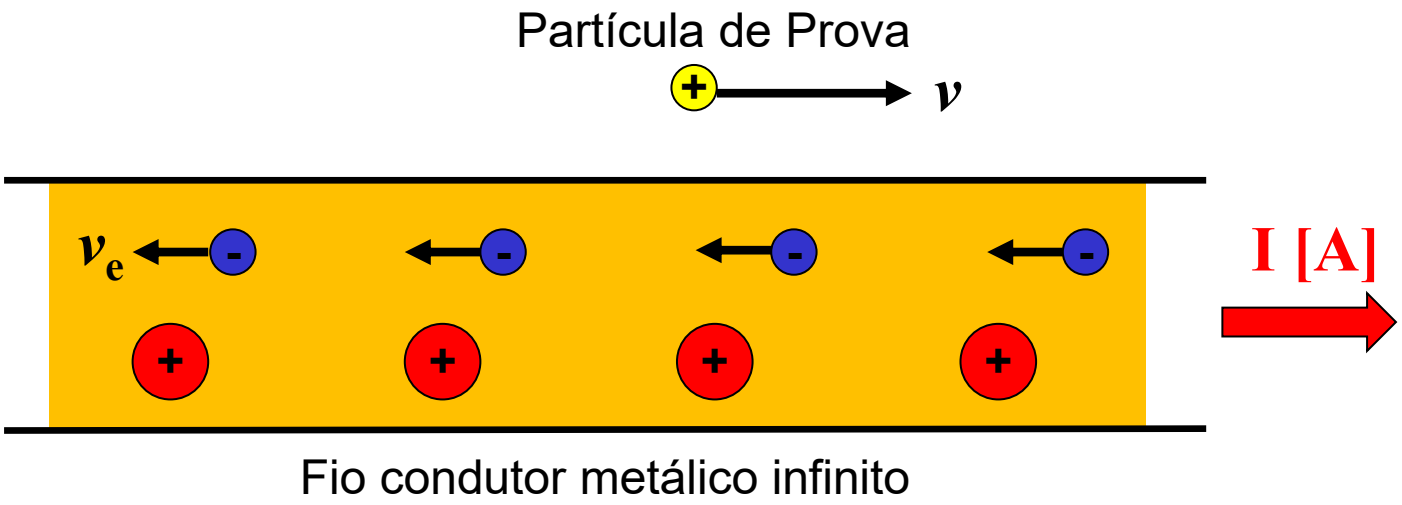
$$\lambda_{rel} = \gamma(v)\lambda_+ - \frac{\gamma(u)}{\gamma(v_e)}\lambda_-$$

$$\lambda_{rel} = \gamma(v)\lambda_+ - \gamma(v)\lambda_-(1 + (vv_e)/c^2)$$

Observada pela Partícula de Prova

Densidade Relativística de Cargas em Condutor

Velocidades Paralelas



Adição de Velocidades [m/s]:

$$u = v + v_e$$

(Galileo)

$$u = (v + v_e)/(1 + (vv_e)/c^2)$$

(Einstein)

Densidade Linear de Cargas [C/m]:

$$\lambda = \lambda_+ - \lambda_- = 0$$

(Repouso)

$$\lambda_{rel} = \gamma(v)\lambda_+ - \gamma(v)\lambda_-(1 + (vv_e)/c^2)$$

$$\boxed{\lambda_{rel} = -\gamma(v)\lambda_- \ (vv_e)/c^2}$$

(Densidade Relativística)

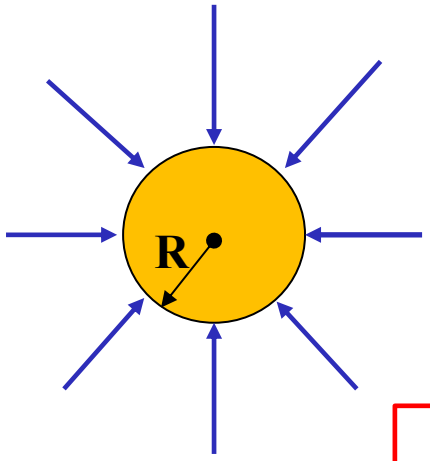
Natureza Relativística dos Campos Magnéticos

Densidade Linear de Cargas [C/m]: $\boxed{\lambda_{rel} = -\gamma(v)\,\lambda_{-}\,(vv_e)/c^2}$

Observada pela Partícula de Prova

Campo Elétrico Radial [V/m]:

$$\boldsymbol{E}_{rad} = \frac{\lambda}{2\pi\varepsilon_0 r} = \frac{-\gamma(v)\lambda_{-}(vv_e)/c^2}{2\pi\varepsilon_0 r} = \frac{-\gamma(v)v\overbrace{(\lambda_{-}v_e)}^{\text{I [A] ou [C/s]}}}{2\pi\varepsilon_0 r c^2}$$



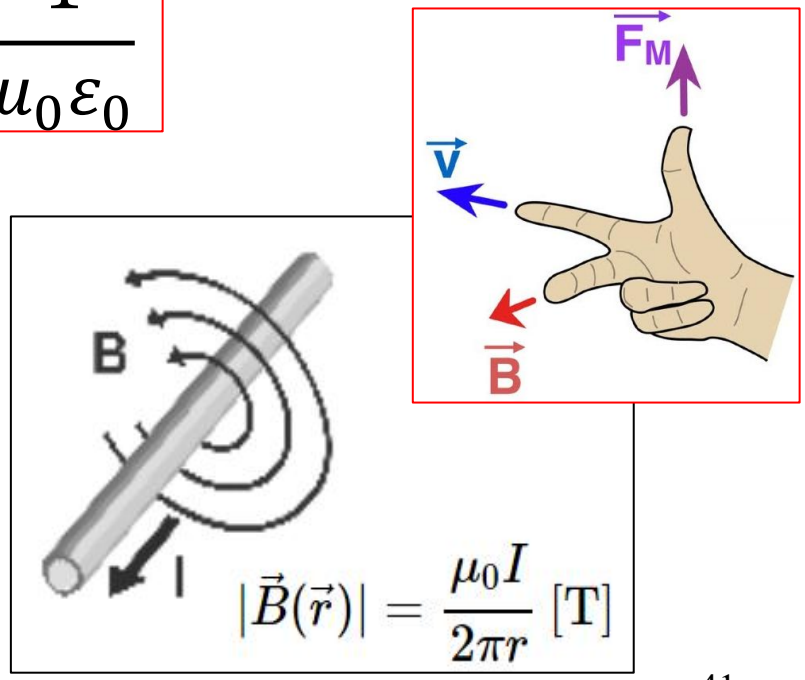
$\boxed{r >> \mathbf{R}}$

$$\boxed{c^2 = \frac{1}{\mu_0\varepsilon_0}}$$

$$\boxed{\boldsymbol{F}_{rad} = \frac{q\boldsymbol{E}_{rad}}{\gamma(v)} = \frac{-q\mu_0 I}{2\pi r}v}$$

Equivalência!

$$\gamma(v) = \frac{1}{\sqrt{1 - v^2/c^2}}$$



$$F_m = q\vec{v} \times \vec{B}$$

Equações da Mecânica Quântica

Planck: $E = h\nu = \hbar\omega$ **Louis de Broglie:** $p = \frac{h}{\lambda} = \hbar k$

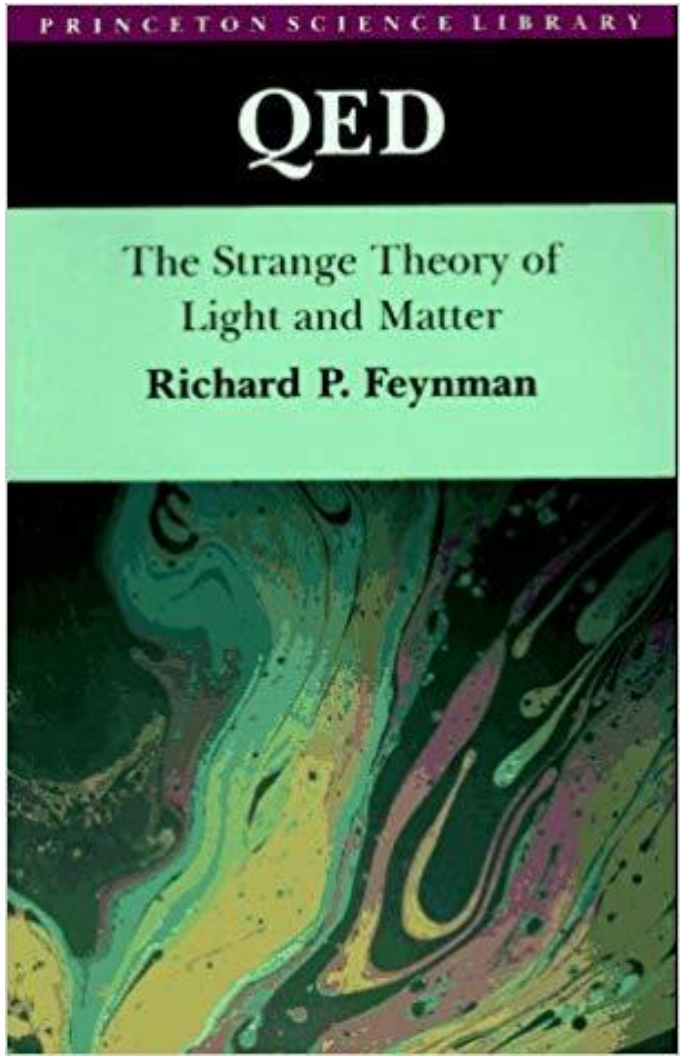
Schrödinger: $i\hbar\frac{\partial}{\partial t}\Psi(x, y, z, t) = -\frac{\hbar^2}{2m}\nabla^2\Psi(x, y, z, t) + V(x)\Psi(x, y, z, t)$

$$i\hbar\frac{\partial}{\partial t}\Psi(x_1, ..., x_n, t) = \hbar^2(-\frac{\nabla_1^2}{2m_1} - \frac{\nabla_2^2}{2m_2} ... - \frac{\nabla_N^2}{2m_N})\Psi(x_1, ..., x_n, t) + V(x_1, .., x_n, t)\Psi(x_1, ..., x_n, t)$$

Dirac: $$\left(\beta mc^2 + \sum_{k=1}^{3}\alpha_k p_k c\right)\psi(\mathbf{x}, t) = i\hbar\frac{\partial\psi(\mathbf{x}, t)}{\partial t}$$

Klein-Gordon: $$-\frac{1}{c}\frac{\partial^2}{\partial t^2}\psi = -\nabla^2\psi + \frac{m^2c^2}{\hbar^2}\psi$$

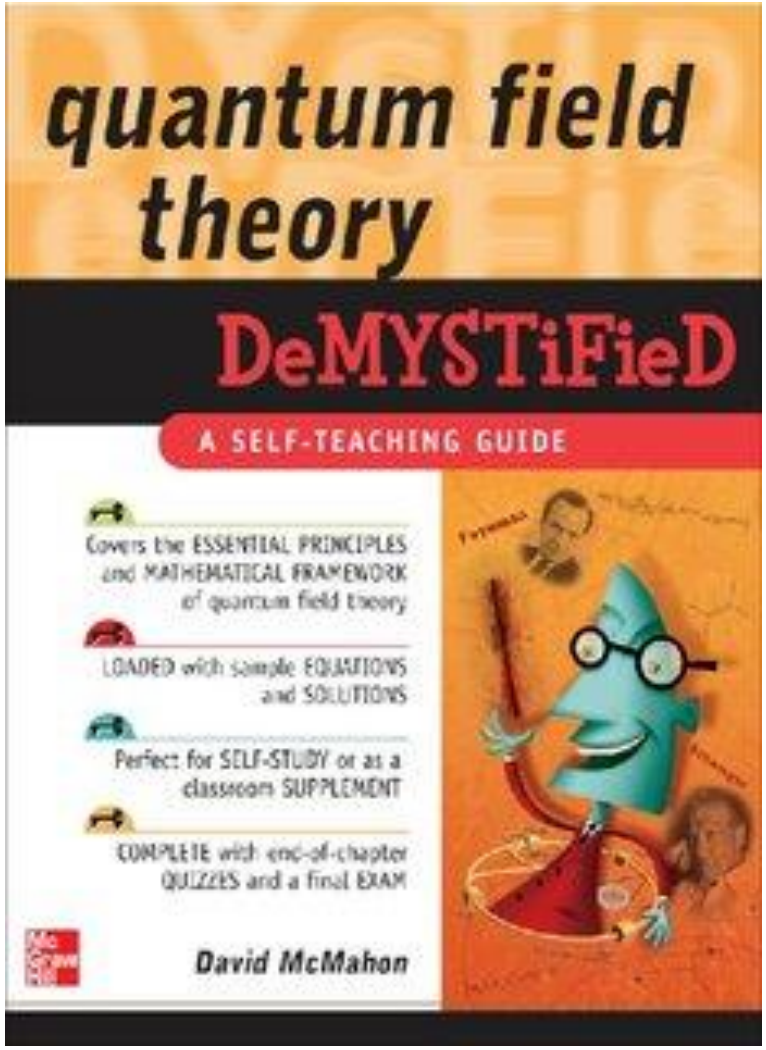
Quantum Electrodynamics



Interação da Matéria com Fótons
(Relatividade Especial)

https://www.amazon.com/QED-Strange-Theory-Light-Matter/dp/0691024170

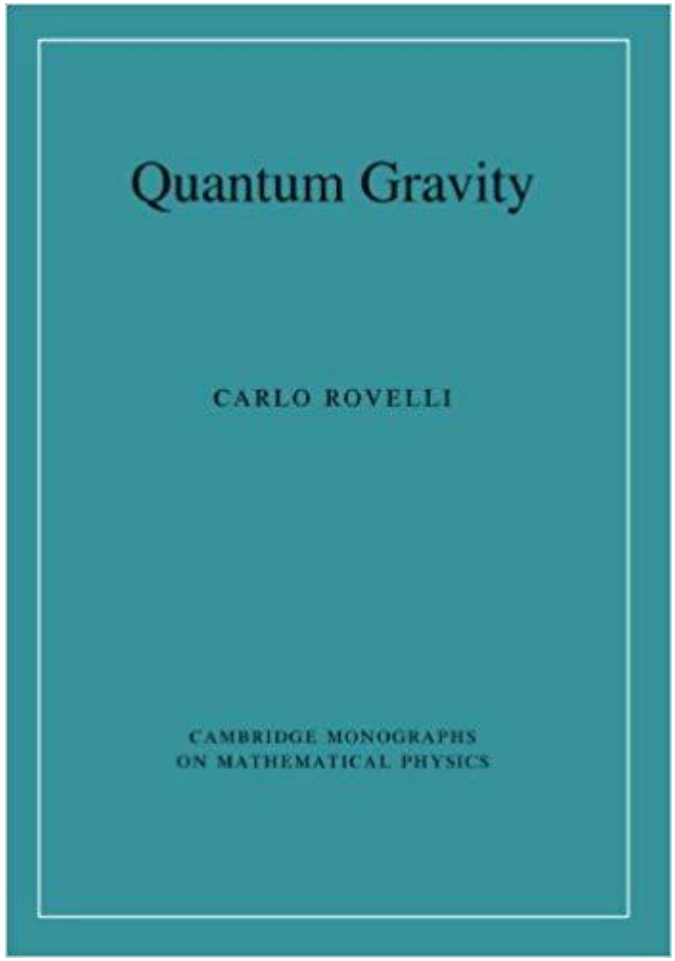
Quantum Field Theory



RELATIVIDADE ESPECIAL + MECÂNICA QUÂNTICA
(Sem a Gravidade!)

https://library.uoh.edu.iq/admin/ebooks/93748-quantum-field-theory-demystified~tqw~_darksiderg.pdf

Quantum Gravity



RELATIVIDADE GERAL + MECÂNICA QUÂNTICA
(Com a Gravidade!)

https://www.amazon.com/Quantum-Gravity-Cambridge-Monographs-Mathematical/dp/0521715962

https://chatgpt.com/

Define what is "The Scientific Method".

The **Scientific Method** is a systematic process used by scientists and researchers to investigate phenomena, acquire new knowledge, or correct and integrate previous knowledge. It involves a series of steps that help ensure that conclusions drawn from experiments and observations are logical, reliable, and objective.

Here are the key steps typically involved in the scientific method:

1. **Observation:** Identifying a problem or an interesting phenomenon that needs explanation.
2. **Question:** Asking a specific, testable question based on the observation.
3. **Hypothesis:** Formulating a hypothesis, which is a tentative explanation or prediction that can be tested through experiments.
4. **Experiment:** Designing and conducting experiments to test the hypothesis. This step involves collecting data through controlled and reproducible methods.
5. **Analysis:** Analyzing the data collected from the experiments to see if they support or contradict the hypothesis.
6. **Conclusion:** Drawing conclusions based on the analysis of the data. If the data support the hypothesis, it may be accepted or refined. If the data contradict the hypothesis, it may be rejected or revised.
7. **Communication:** Sharing the results and conclusions with the scientific community, often through publications or presentations, allowing for peer review and further investigation.
8. **Replication:** Repeating the experiments to verify results and ensure reliability. Replication helps confirm that findings are consistent and not due to errors or biases.

The scientific method is a dynamic process, often requiring revisions and repeated experiments to refine hypotheses and build deeper understanding.

https://chatgpt.com/

Generate an image of a beautifull sunset

Created with DALL·E

Here's a beautiful sunset over the ocean, with warm colors blending in the sky and a golden reflection

Top Generative AI Chatbots by Market Share – March 2026

Rank	Generative AI ChatBot	Description	Included LLMs	AI Search Market Share	Estimated Quarterly User Growth
1	ChatGPT (excluding Copilot)	General-purpose AI chatbot	GPT-5.2, GPT-5, GPT-4.1	60.4%	4% ▲
2	Google Gemini	General-purpose AI assistant	Gemini 3 family	15.2%	12% ▲
3	Microsoft Copilot	General-purpose AI assistant	GPT-5.2, GPT-5.1, Microsoft Prometheus	12.9%	3% ▲
4	Perplexity	Accuracy-focused AI search engine	Sonar, Sonar Pro	5.8%	4% ▲
5	Claude AI	Business-focused AI assistant	Claude Sonnet 4.5, Claude Haiku 4.5, Opus 4	4.5%	14% ▲
6	Grok	General-purpose AI search engine	Grok 4.1	0.6%	4% ▲
7	Deepseek	General-purpose AI search engine	DeepSeek V3.2, DeepSeek R1	0.2%	7% ▲
8	Brave Leo AI	Privacy-focused AI assistant	Mixtral 8×7B, Llama 3.1	0.1%	3% ▲
9	Komo	Link-surfacing AI search engine	Not publicly disclosed	0.1%	2% ▲
10	Andi	Simplicity-focused AI search engine	Not publicly disclosed	0.1%	4% ▲

https://firstpagesage.com/reports/top-generative-ai-chatbots/

https://chatgpt.com/

ChatGPT

Generate C code to print a list of prime numbers up to 1000

Here's a simple C program that prints all prime numbers up to 1000.

Prime Numbers

Responder no chat >

This C program defines a function to check for prime numbers and prints all primes up to 1000. Let me know if you need any modifications!

Esta conversa foi útil até agora?

Prime Numbers

```c
#include <stdio.h>

// Function to check if a number is prime
int is_prime(int num) {
    if (num < 2) return 0;
    for (int i = 2; i * i <= num; i++) {
        if (num % i == 0) return 0;
    }
    return 1;
}

int main() {
    printf("Prime numbers up to 1000:\n");
    for (int i = 2; i <= 1000; i++) {
        if (is_prime(i)) {
            printf("%d ", i);
        }
    }
    printf("\n");
    return 0;
}
```

main.c

```c
#include <stdio.h>

// Function to check if a number is prime
int is_prime(int num) {
    if (num < 2) return 0;
    for (int i = 2; i * i <= num; i++) {
        if (num % i == 0) return 0;
    }
    return 1;
}

int main() {
    printf("Prime numbers up to 1000:\n");
    for (int i = 2; i <= 1000; i++) {
        if (is_prime(i)) {
            printf("%d ", i);
        }
    }
    printf("\n");
    return 0;
}
```

```
Prime numbers up to 1000:
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89
 97 101 103 107 109 113 127 131 137 139 149 151 157 163 167 173 179
 181 191 193 197 199 211 223 227 229 233 239 241 251 257 263 269 27
1 277 281 283 293 307 311 313 317 331 337 347 349 353 359 367 373 3
79 383 389 397 401 409 419 421 431 433 439 443 449 457 461 463 467
479 487 491 499 503 509 521 523 541 547 557 563 569 571 577 587 593
 599 601 607 613 617 619 631 641 643 647 653 659 661 673 677 683 69
1 701 709 719 727 733 739 743 751 757 761 769 773 787 797 809 811 8
21 823 827 829 839 853 857 859 863 877 881 883 887 907 911 919 929
937 941 947 953 967 971 977 983 991 997
```

https://www.onlinegdb.com/online_c_compiler

1ª Pontuação Extra [até 1,0 ponto]

Escolha um Chatbot de Inteligência Artificial Generativa.
Faça um teste com perguntas ou pedidos para obter um erro óbvio.

Dicas:
- IAs geralmente não são eficazes na solução de problemas que envolvem matemática e lógica.

- Muitas **IAs** comerciais tendem a concordar de forma complacente com as afirmações erradas dos usuários. Quando desafiadas, algumas **IAs** comerciais tendem a assumir erros onde estes não existem.

- **IAs** podem alucinar.

Comente e faça críticas sobre o resultado obtido (Obrigatório!)

Inclua referências para justificar os seus comentários e críticas.

Entrega: Até 24/03/26 (terça-feira)
Máximo de 1 página impressa em papel.
Não serão aceitas folhas manuscritas.